Multivariate and Velocity-Resolved Reverberation in NGC 5548: Astrophysical Motivation for an Information-Based Analysis of Continuum–BLR Coupling

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Abstract

Reverberation mapping has become one of the most powerful tools for probing the unresolved inner structure of active galactic nuclei. In its classical form, the method relates continuum variability to delayed emission-line responses and thereby constrains the characteristic scale of the broad-line region, the black-hole mass, and the geometry and kinematics of the line-emitting gas. However, modern high-cadence campaigns have shown that the astrophysical situation is richer than the traditional single-driver, single-response picture suggests. In well-observed systems such as NGC 5548, continuum interband delays, velocity-resolved Hβ reverberation, and episodes of anomalous line–continuum decorrelation indicate that the observed optical or ultraviolet continuum is not always an adequate representation of the ionizing field seen by the broad-line region. These results motivate a more explicitly multivariate and velocity-resolved analysis of reverberation data, especially under conservative time-domain conditions that minimize artificial structure introduced by incomplete temporal overlap. In this paper we focus on the astrophysical problem: how the optical continuum and the blue wing, core, and red wing of Hβ jointly encode the dynamical response of the low-ionization broad-line region in NGC 5548. The information-decomposition formalism employed later in the paper is not introduced here; instead, this Introduction develops the astrophysical background, motivates the scientific questions, and places the present work in the context of the reverberation-mapping literature.

1 Introduction

Active galactic nuclei (AGN) are powered by accretion onto supermassive black holes and radiate over a broad range of wavelengths through a compact but highly structured central engine. The standard observational picture includes an accretion disk that produces much of the optical and ultraviolet continuum, a hot corona responsible for the X-ray emission, a broad-line region (BLR) composed of dense photoionized gas moving at velocities of order 10^3 to 10^4 km s^{-1}, and more distant dusty material that reprocesses the radiation field at infrared wavelengths [**Peterson2004**, **Netzer2013**, **Cackett2021**]. Because these regions are too compact to resolve directly in most AGN, temporal variability provides one of the few effective ways to infer their sizes, structure, and coupling.

Reverberation mapping exploits the fact that fluctuations in the central ionizing source are echoed by delayed responses in remote reprocessing or emission-line regions. In its simplest linearized form, the response of an emission line $L(t)$ to a driving continuum $C(t)$ may be written as

$$L(t) = \int \Psi(\tau)\, C(t-\tau)\, d\tau, \tag{1}$$

where $\Psi(\tau)$ is the transfer function or delay distribution. The classical use of this formalism has been to derive a characteristic lag τ, associate it with a BLR radius $R \sim c\tau$, and combine that scale with the line width to estimate the black-hole mass under a virial assumption [**Peterson2004**, **BentzKatz2015**]. This framework has been extremely successful and remains foundational to AGN black-hole mass scaling relations.

Nevertheless, the astrophysical interpretation of reverberation data has become progressively more subtle. The quantity that drives the broad emission lines is the ionizing continuum, particularly in the extreme ultraviolet and soft X-ray bands, whereas the continuum actually monitored in many optical reverberation campaigns is a more accessible proxy, often the flux near 5100 Å. As a result, the measured lag between an observed continuum band and an emission line need not correspond to a direct causal connection between those two observables alone. Instead, both may be responding, with different delays and different filtering, to a latent radiation field that is only imperfectly traced by the observed bands [**Cackett2021**]. The question is therefore not merely what the lag is, but whether the monitored continuum contains the relevant information needed to explain the line response.

This issue is underscored by continuum reverberation mapping. A series of high-cadence monitoring programs has revealed that ultraviolet and optical continuum bands in AGN are delayed with respect to one another, with delays that often grow systematically with wavelength as expected for a thermally stratified accretion disk. However, the measured lags are frequently larger than predicted by the simplest geometrically thin, optically thick accretion-disk models [**Fausnaugh2016**, **Fausnaugh2017**, **Edelson2019**, **Cackett2021**]. These results suggest that the observed continuum variability is not fully described by a minimal reprocessing picture and may involve additional processes such as intrinsic accretion-rate fluctuations, diffuse continuum emission, complex irradiation geometry, or nontrivial radiative transfer. From the standpoint of BLR reverberation, this implies that optical continuum light curves may not be interpreted as pristine drivers without further caution.

NGC 5548 is the canonical source in which these concerns become impossible to ignore. It is one of the most intensively monitored Seyfert galaxies and has served as a cornerstone of reverberation mapping for decades. It has played a major role in calibrating broad-line lags, virial mass estimates, and radius–luminosity relations, while more recent campaigns have transformed it into a benchmark for multiwavelength and velocity-resolved reverberation analysis [**Peterson2002**, **BentzKatz2015**, **Cackett2021**]. In particular, the AGN STORM campaign delivered one of the richest reverberation datasets ever obtained, including optical and ultraviolet continuum monitoring, extensive spectroscopy, and detailed velocity-delay information for several broad lines [**DeRosa2015**, **Pei2017**, **Horne2021**].

These observations revealed that NGC 5548 is not well described by a stationary one-zone BLR driven by a single observed continuum proxy. Velocity-resolved reverberation results showed clear structure across the Hβ profile, with the line wings and core participating differently in the reverberation pattern [**Pei2017**, **Horne2021**]. The interpretation of these data points toward a BLR that is radially stratified and kinematically complex, possibly including multiple components or zones rather than a single homogeneous distribution of clouds [**Horne2021**, **LongDexter2025**]. This immediately motivates a shift away from purely integrated-line analyses. If the blue wing, line core, and red wing carry different dynamical information, then treating Hβ as a single scalar time series necessarily compresses part of the astrophysical content of the data.

An even more striking complication emerged during the same source's monitoring history: the so-called "holiday" in NGC 5548. During part of the AGN STORM campaign, the broad emission lines became partially decorrelated from the observed UV continuum, even though they had previously tracked it closely. The leading interpretation is that an obscuring wind or shielding structure modified the ionizing spectral energy distribution reaching the BLR while leaving the observed UV continuum comparatively unaffected [**Dehghanian2019**]. The importance of this result cannot be overstated. It demonstrates directly that an observed continuum channel may cease to be a faithful tracer of the radiation field relevant for BLR photoionization. Consequently, a line–continuum lag analysis may look physically incomplete not because the line has stopped reverberating, but because the monitored continuum is no longer the appropriate driver proxy.

This broader lesson is now reinforced by work across the reverberation-mapping literature. Velocity-resolved Hβ studies increasingly show that BLR kinematics are diverse across AGN and may evolve over time, with some sources displaying signatures consistent with inflow, others with outflow, and others with virialized or mixed dynamics [**Du2018**, **Wang2025**, **Feng2024**]. In parallel, changing-look AGN have demonstrated that the relation between continuum state and broad-line emission can undergo dramatic reorganization on humanly accessible timescales [**RicciTrakhtenbrot2023**]. Although changing-look phenomena are not the focus of the present study, they help establish a wider astrophysical principle: the mapping between observed continuum, ionizing continuum, and line-emitting gas is not guaranteed to be stationary.

Within this context, NGC 5548 is particularly well suited for a more conservative, multivariate analysis of the reverberation problem. The source offers high-quality continuum monitoring, extensive Hβ spectroscopy, and a rich legacy of prior reverberation results against which new analyses may be compared. At the same time, it is sufficiently complex to require such an approach. If one asks only for an integrated Hβ lag relative to the optical continuum, much of the internal structure of the reverberation process is averaged away. By contrast, if one treats the blue wing, core, and red wing of Hβ as distinct observables, new astrophysical questions become accessible. Does the core behave primarily as an echo of the continuum, or does it also depend on information already present in the wings? Are the two wings symmetric in their predictive role, as one might expect in an idealized virialized BLR, or does one side of the profile carry systematically different dynamical information? Can the observed optical continuum alone account for the velocity-resolved response, or do hidden drivers and internal line-component couplings remain necessary?

These questions are not purely methodological. They arise naturally from longstanding astrophysical uncertainties about the BLR and the continuum source. The BLR is expected to be sensitive to the ionizing continuum, but in practice the ionizing continuum is usually unobserved. The optical continuum is measurable but is itself part of a reverberating accretion-flow system and may therefore contain both intrinsic and reprocessed variability components [**Fausnaugh2017**, **Cackett2021**]. The Hβ line is observable with high fidelity, but its integrated flux merges together multiple kinematic regions whose relative responses may differ. Once velocity-resolved spectroscopy is available, the astrophysical challenge becomes one of causal disentanglement among observables that are neither independent nor trivially ordered in a simple driver–response chain.

A number of established approaches address parts of this challenge. Interpolated cross-correlation methods and related lag estimators remain the standard tools for continuum–line reverberation analysis and continue to provide essential baseline measurements [**Peterson2004**, **WhitePeterson1994**]. JAVELIN and related stochastic-process approaches model reverberation as a damped random walk continuum filtered by a transfer function, thereby improving lag estimation and uncertainty quantification in many cases [**Zu2011**, **Zu2013**]. Velocity-delay mapping and dynamical modeling aim to reconstruct the geometry and kinematics of the BLR directly from the line-profile response [**Pancoast2014**, **Horne2021**]. Continuum reverberation studies focus on interband disk lags and the thermal structure of the accretion flow [**Fausnaugh2016**, **Fausnaugh2017**, **Edelson2019**]. Each of these approaches has been productive, but they are typically tailored to a specific projection of the problem: pairwise lags, transfer-function fitting, dynamical inversion, or continuum size scaling.

What remains comparatively underexplored is the specific astrophysical question of how multiple simultaneously observed channels jointly constrain the future evolution of one chosen component of the system, especially when those channels include both the continuum and velocity-resolved line segments. In other words, once the Hβ profile has been decomposed into blue wing, core, and red wing light curves, and once a conservative common time interval is imposed, the natural problem is no longer simply "what is the lag?" It becomes: how much of the future behavior of one velocity component is already encoded in the continuum, how much is encoded in another part of the line profile, and how much lies beyond the information available in the monitored observables? That is the astrophysical problem motivating the present paper.

The emphasis on strict temporal overlap is also not incidental. Irregular cadence, seasonal gaps, and unequal overlap among observed channels are intrinsic features of AGN monitoring data. Any multivariate time-series analysis is therefore vulnerable to spurious structure introduced by aggressive interpolation, extrapolation across gaps, or implicit assumptions about the simultaneity of channels that were not truly observed together. In the context of NGC 5548, where the scientific questions depend sensitively on comparing the optical continuum to distinct Hβ velocity components, it is essential to work under conservative overlap conditions. The goal is not to maximize sample size at all costs, but to preserve physical interpretability. A shorter but genuinely shared time interval is preferable to a longer one in which apparent multivariate structure may be partly synthetic.

The present paper is therefore built around a deliberately astrophysical motivation. We seek to determine how the optical continuum near 5100 Å and the velocity-resolved components of Hβ in NGC 5548 are coupled when the data are restricted to robust common overlap. Our interest is not merely in recovering a known lag, but in characterizing the continuum–BLR system as a multichannel, time-dependent structure. We focus on the blue wing, line core, and red wing because these components are directly linked to the low-ionization BLR geometry and kinematics, and because they allow one to ask whether different parts of the line profile participate differently in the reverberation process. This is the astrophysical motivation for the analysis developed in the sections that follow.

The methodological framework used later in the paper is based on information decomposition, but we intentionally postpone its formal description. The present Introduction has a different role: to show why such an analysis is astrophysically warranted. The literature has already established that NGC 5548 exhibits continuum interband lags, velocity-resolved Hβ structure, partial line–continuum decoupling during the holiday, and evidence for a multicomponent BLR. The natural next step is not simply to add another lag estimator, but to investigate the reverberation data in a way that is explicitly multivariate, velocity-resolved, and conservative with respect to temporal overlap. That is the problem we address here.

The structure of this paper is as follows. In Section 2 we describe the observations, spectral processing, and the construction of strictly overlapping continuum and velocity-resolved Hβ time series. In Section 3 we introduce the formal information-based framework used to quantify the coupling among the observables. In Section 4 we present the main results for continuum–line and line–line relationships, including robustness tests and comparisons with classical reverberation measures. In Section 5 we discuss the implications for the structure of the BLR in NGC 5548, the role of hidden or imperfectly observed drivers, and the interpretation of multivariate reverberation signals. Section 6 summarizes our conclusions and outlines possible extensions to larger AGN samples and to other broad emission lines.

2 Synergistic–Unique–Redundant Decomposition (SURD) for Time Series

2.1 Problem formulation

Consider a collection of N observed stochastic processes,

$$\mathbf{Q}(t) = \big(Q_1(t), Q_2(t), \ldots, Q_N(t)\big), \tag{2}$$

sampled at discrete times $t = t_n$. For a chosen target index j, the goal is to quantify how much information about the future state

$$Q_j^+ \equiv Q_j(t + \Delta T) \tag{3}$$

is contained in the observed variables at time t and, crucially, to decompose that predictive information into *redundant*, *unique*, and *synergistic* contributions [**MartinezSanchez2024SURD**].

The central motivation is that in multivariate systems, pairwise predictive measures such as Granger causality or transfer entropy can establish that a variable is informative about a target, but they do not in general distinguish whether that information is:

1. unique to one predictor,
2. duplicated across several predictors, or
3. available only from a joint observation of several predictors.

SURD is designed precisely to perform that decomposition in a time-series setting [**MartinezSanchez2024SURD**, **Schreiber2000**, **WilliamsBeer2010**].

2.2 Information-theoretic preliminaries

Let $Y \equiv Q_j^+$ denote the target and let $\mathbf{X}$ denote the vector of observed predictors. For discrete variables, the Shannon entropy is

$$H(Y) = -\sum_y p(y) \log_2 p(y), \tag{4}$$

and the conditional entropy of Y given $\mathbf{X}$ is

$$H(Y \mid \mathbf{X}) = -\sum_{y,\mathbf{x}} p(y, \mathbf{x}) \log_2 p(y \mid \mathbf{x}). \tag{5}$$

The mutual information between Y and $\mathbf{X}$ is

$$I(Y; \mathbf{X}) = H(Y) - H(Y \mid \mathbf{X}) = \sum_{y,\mathbf{x}} p(y, \mathbf{x}) \log_2 \frac{p(y, \mathbf{x})}{p(y)p(\mathbf{x})}. \tag{6}$$

This quantity measures the total predictive information about the future target that is available in the observed present state [**Shannon1948**, **CoverThomas2006**].

In SURD, the unexplained part of the future is interpreted as a *causality leak*,

$$\Delta I_{\text{leak}\to j} = H(Y \mid \mathbf{X}), \tag{7}$$

so that the forward information balance reads

$$H(Y) = I(Y; \mathbf{X}) + \Delta I_{\text{leak}\to j}. \tag{8}$$

Equation (8) is one of the core ideas of SURD: the future uncertainty of the target splits into an observed part, explained by the measured variables, and an unobserved part, attributed to hidden or unmeasured influences [**MartinezSanchez2024SURD**].

2.3 From total predictive information to causal components

Let the predictor vector be

$$\mathbf{X} = (X_1, \ldots, X_M), \tag{9}$$

where the X_α may correspond either to different physical variables at the same time, or to different lagged copies of the same variables. SURD decomposes the observable predictive information $I(Y; \mathbf{X})$ into additive components:

$$I(Y; \mathbf{X}) = \sum_{\mathcal{A} \in \mathcal{C}} \Delta I^R_{\mathcal{A} \to j} + \sum_{\alpha=1}^{M} \Delta I^U_{\alpha \to j} + \sum_{\mathcal{A} \in \mathcal{C}} \Delta I^S_{\mathcal{A} \to j}, \tag{10}$$

where:

- $\Delta I^R_{\mathcal{A} \to j}$ is the *redundant* contribution from the group of predictors indexed by $\mathcal{A}$,
- $\Delta I^U_{\alpha \to j}$ is the *unique* contribution of predictor X_α,
- $\Delta I^S_{\mathcal{A} \to j}$ is the *synergistic* contribution from the group $\mathcal{A}$,
- $\mathcal{C}$ denotes all non-singleton subsets of $\{1, \ldots, M\}$.

The full future entropy therefore obeys

$$H(Y) = \sum_{\mathcal{A} \in \mathcal{C}} \Delta I^R_{\mathcal{A} \to j} + \sum_{\alpha=1}^{M} \Delta I^U_{\alpha \to j} + \sum_{\mathcal{A} \in \mathcal{C}} \Delta I^S_{\mathcal{A} \to j} + \Delta I_{\mathrm{leak} \to j}. \tag{11}$$

The three causal types have the following interpretation [**MartinezSanchez2024SURD**]:

- Redundant: the same predictive information is shared across several predictors,

- Unique: a predictor contains information not available from any other predictor alone,

- Synergistic: information appears only when several predictors are observed jointly.

2.4 Specific mutual information and statewise decomposition

A key feature of SURD is that the source of predictability may depend on which *particular future event* y occurs. To resolve this dependence, SURD works first at the level of *specific mutual information* [**DeWeeseMeister1999**, **Butts2003**, **MartinezSanchez2024SURD**].

For a particular target state $Y = y$, the specific mutual information between y and a predictor set $\mathbf{X}_\mathcal{A}$ is

$$\tilde{\imath}_\mathcal{A}(y) \equiv \tilde{\imath}\big(y; \mathbf{X}_\mathcal{A}\big) = \sum_{\mathbf{x}_\mathcal{A}} \frac{p(y, \mathbf{x}_\mathcal{A})}{p(y)} \log_2 \frac{p(y, \mathbf{x}_\mathcal{A})}{p(y)\, p(\mathbf{x}_\mathcal{A})}. \tag{12}$$

The ordinary mutual information is recovered by averaging over target states:

$$I\big(Y; \mathbf{X}_\mathcal{A}\big) = \sum_y p(y)\, \tilde{\imath}_\mathcal{A}(y). \tag{13}$$

The conceptual reason for introducing $\tilde{\imath}_\mathcal{A}(y)$ is that different predictors may be informative for different target outcomes. A predictor may, for example, strongly constrain large positive target excursions while another may mainly constrain negative or low-amplitude states. A decomposition performed only at the level of the average mutual information can obscure this state dependence. SURD therefore computes the information contributions for each y, decomposes them into redundant, unique, and synergistic increments, and only afterwards averages over y [**MartinezSanchez2024SURD**].

2.5 Statewise incremental construction of SURD

Let $\mathfrak{P}^*(\{1, \ldots, M\})$ denote the set of all non-empty subsets of predictors. For each subset $\mathcal{A} \in \mathfrak{P}^*(\{1, \ldots, M\})$, one computes

$$\tilde{\imath}_\mathcal{A}(y) = \tilde{\imath}\big(y; \mathbf{X}_\mathcal{A}\big). \tag{14}$$

SURD then considers the collection of all these values for a fixed target state y and identifies the *increments* in predictive information that arise as one moves from smaller to larger subsets of predictors [**MartinezSanchez2024SURD**]. Denoting these statewise increments schematically by

$$\Delta\tilde{\imath}_\mathcal{A}(y), \tag{15}$$

the classification rule is:

- an increment assigned to one predictor alone contributes to $\Delta\tilde{\imath}^{U}_{\alpha}(y)$,
- an increment shared across several predictors contributes to $\Delta\tilde{\imath}^{R}_{\mathcal{A}}(y)$,
- an increment that arises only when a collection of predictors is observed jointly contributes to $\Delta\tilde{\imath}^{S}_{\mathcal{A}}(y)$.

The averaged causal components are then

$$\Delta I^{R}_{\mathcal{A}\to j} = \sum_{y} p(y)\,\Delta\tilde{\imath}^{R}_{\mathcal{A}}(y), \tag{16}$$

$$\Delta I^{U}_{\alpha\to j} = \sum_{y} p(y)\,\Delta\tilde{\imath}^{U}_{\alpha}(y), \tag{17}$$

$$\Delta I^{S}_{\mathcal{A}\to j} = \sum_{y} p(y)\,\Delta\tilde{\imath}^{S}_{\mathcal{A}}(y). \tag{18}$$

This construction ensures that the total observable predictive information is recovered exactly:

$$I(Y;\mathbf{X}) = \sum_{\mathcal{A}\in\mathcal{C}} \Delta I^{R}_{\mathcal{A}\to j} + \sum_{\alpha=1}^{M} \Delta I^{U}_{\alpha\to j} + \sum_{\mathcal{A}\in\mathcal{C}} \Delta I^{S}_{\mathcal{A}\to j}. \tag{19}$$

2.6 Two-source case

For two predictors X_1 and X_2, the decomposition reduces to

$$I(Y;X_1,X_2) = \Delta I^{R}_{12\to j} + \Delta I^{U}_{1\to j} + \Delta I^{U}_{2\to j} + \Delta I^{S}_{12\to j}. \tag{20}$$

Consistency with the single-predictor mutual informations requires

$$I(Y;X_1) = \Delta I^{R}_{12\to j} + \Delta I^{U}_{1\to j}, \tag{21}$$

$$I(Y;X_2) = \Delta I^{R}_{12\to j} + \Delta I^{U}_{2\to j}. \tag{22}$$

Hence the synergistic contribution may be written as the remainder

$$\Delta I^{S}_{12\to j} = I(Y;X_1,X_2) - \Delta I^{R}_{12\to j} - \Delta I^{U}_{1\to j} - \Delta I^{U}_{2\to j}. \tag{23}$$

This form makes the interpretation transparent: the synergy is the part of the joint predictive information that cannot be accounted for by the redundant and unique terms.

2.7 Extension to time series with multiple lags

For time-series applications, the predictor vector is typically augmented with lagged copies of the observed variables. If one includes p past lags per variable, the predictor vector becomes

$$\mathbf{X}(t) = \Big[Q_1(t), Q_1(t-\Delta T_1), \ldots, Q_1(t-\Delta T_p),\ Q_2(t), Q_2(t-\Delta T_1), \ldots, Q_N(t-\Delta T_p)\Big]. \tag{24}$$

For example, with $N = 2$ variables and one additional lag,

$$\mathbf{X}(t) = \big[Q_1(t), Q_1(t-\Delta T), Q_2(t), Q_2(t-\Delta T)\big]. \tag{25}$$

SURD then treats each lagged component as one predictor channel and applies the same decomposition machinery to the enlarged predictor set [**MartinezSanchez2024SURD**]. In this way, the method can distinguish:

- self-causation from $Q_j(t-\Delta T_k)$ to $Q_j(t+\Delta T)$,
- cross-causation from $Q_i(t-\Delta T_k)$ to $Q_j(t+\Delta T)$,
- redundancy between different lags,
- synergy between variables and lags.

This is particularly important when several lagged channels may encode overlapping predictive content. For example, adding $Q_j(t)$ and $Q_j(t-\Delta T)$ can increase redundancy rather than revealing new independent causal pathways. SURD is expressly designed to make that distinction [**MartinezSanchez2024SURD**].

2.8 Contemporaneous and mixed lagged dependencies

A further advantage of the formalism is that the predictor vector can include variables at time $t + \Delta T$ other than the target itself, thereby allowing the analysis of contemporaneous or effectively sub-resolution interactions [**MartinezSanchez2024SURD**]. In practice, one excludes the target variable from the contemporaneous predictor set to avoid trivial self-determination, but permits other variables measured on the same time slice. Thus, a generic predictor vector may mix present, lagged, and contemporaneous channels:

$$\mathbf{X}(t) = \big[\text{past lags}, \text{present variables}, \text{contemporaneous non-target variables}\big]. \tag{26}$$

The same decomposition (11) then applies.

2.9 Normalization and interpretation

Because the observable predictive information is conserved by construction, the unique, redundant, and synergistic terms are naturally normalized by the total observable mutual information:

$$\widehat{\Delta I}^{R}_{\mathcal{A}\to j} = \frac{\Delta I^{R}_{\mathcal{A}\to j}}{I(Y;\mathbf{X})}, \qquad \widehat{\Delta I}^{U}_{\alpha\to j} = \frac{\Delta I^{U}_{\alpha\to j}}{I(Y;\mathbf{X})}, \qquad \widehat{\Delta I}^{S}_{\mathcal{A}\to j} = \frac{\Delta I^{S}_{\mathcal{A}\to j}}{I(Y;\mathbf{X})}. \tag{27}$$

These normalized terms sum to unity:

$$\sum_{\mathcal{A}\in\mathcal{C}} \widehat{\Delta I}^{R}_{\mathcal{A}\to j} + \sum_{\alpha=1}^{M} \widehat{\Delta I}^{U}_{\alpha\to j} + \sum_{\mathcal{A}\in\mathcal{C}} \widehat{\Delta I}^{S}_{\mathcal{A}\to j} = 1. \tag{28}$$

The leak is normalized instead by the total entropy of the target:

$$\widehat{\Delta I}_{\mathrm{leak}\to j} = \frac{\Delta I_{\mathrm{leak}\to j}}{H(Y)} = \frac{H(Y \mid \mathbf{X})}{H(Y)}. \tag{29}$$

Hence:

- $\widehat{\Delta I}_{\mathrm{leak}\to j} \approx 0$ means the observed predictors account for almost all predictive information about the target,
- $\widehat{\Delta I}_{\mathrm{leak}\to j} \approx 1$ means most of the target's future uncertainty remains unexplained by the observed variables.

2.10 Practical estimation

In applications, the probabilities appearing in Eqs. (6) and (12) must be estimated from finite data. The original SURD work uses histogram-based and transport-map-based estimators depending on dimensionality and sample size [**MartinezSanchez2024SURD**]. In a histogram implementation, one discretizes the predictor-target space into bins and estimates

$$p(y,\mathbf{x}) \approx \frac{n(y,\mathbf{x})}{N_{\mathrm{samples}}}, \tag{30}$$

where $n(y,\mathbf{x})$ is the bin count. The resulting empirical joint distribution is then used to compute all required marginal and conditional probabilities. More sophisticated estimators, such as transport maps, are advantageous in high dimensions because the number of bins required by direct histograms grows exponentially with the number of predictors.

Algorithmically, the SURD workflow for a target Q_j^+ is:

1. choose the predictor vector $\mathbf{X}$, including variables, lags, or contemporaneous channels;
2. estimate the probability distributions $p(y, \mathbf{x}_{\mathcal{A}})$ for all relevant predictor subsets $\mathcal{A}$;
3. compute the specific informations $\tilde{\imath}_{\mathcal{A}}(y)$ for every target state y;
4. decompose the statewise increments into redundant, unique, and synergistic parts;
5. average over y to obtain ΔI^R, ΔI^U, and ΔI^S;
6. compute the leak from Eq. (7);
7. normalize the observable components by $I(Y; \mathbf{X})$ and the leak by $H(Y)$.

2.11 Properties and assumptions

SURD has several properties that are important in time-series applications [**MartinezSanchez2024**

1. **Non-negativity:** all redundant, unique, synergistic, and leak terms are non-negative.
2. **Additivity:** the observable predictive information is exactly partitioned into R, U, and S terms.
3. **Sensitivity to nonlinear dependence:** because the construction is based on mutual information, it is not restricted to linear couplings.
4. **Compatibility with lagged and contemporaneous predictors:** time-lag structure enters through the predictor vector.
5. **Hidden-variable diagnostics:** the leak quantifies the fraction of future uncertainty left unexplained by the observed variables.

The main assumptions are those common to observational information-theoretic causality analysis:

- the processes are sufficiently stationary over the analysis interval for probability estimation to be meaningful,
- the chosen time resolution is physically relevant,
- the sample size is adequate for the selected probability estimator,
- and causal interpretation remains limited to the observed variables plus the quantified unresolved contribution represented by the leak.

2.12 Summary

SURD starts from the predictive information $I(Q_j^+; \mathbf{X})$ between a future target and a multivariate predictor state. It then decomposes that predictive information into state-averaged unique, redundant, and synergistic contributions, while retaining the conditional entropy $H(Q_j^+ \mid \mathbf{X})$ as an explicit measure of hidden or unobserved influence. Its time-series extension is straightforward: past lags, present variables, and even contemporaneous non-target channels are simply incorporated into the predictor vector. For multivariate dynamical systems, this yields a decomposition that is richer than pairwise lag measures because it distinguishes direct predictive effects from duplicated information and from genuinely joint multivariate effects [**MartinezSanchez2024SURD**, **WilliamsBeer2010**].

3 Observations, Spectral Processing, and Strict Overlap Alignment

To evaluate the capabilities of the SURD algorithm on real-world astrophysical data, we analyze the well-monitored Seyfert 1 galaxy NGC 5548. This source has a rich legacy of reverberation mapping, including the high-cadence AGN STORM campaign [**DeRosa2015**, **Pei2017**]. Our analysis utilizes two main datasets: the optical continuum flux at 5100 Å and velocity-resolved Hβ emission line profiles.

3.1 Data Products and MJD Correction

The 5100 Å continuum light curve is obtained from the AGN Watch databases (specifically, `c5100.dat`), spanning a long duration of several years. The integrated Hβ emission line light curve is similarly taken from clean AGN Watch compilations (`ngc5548_agnwatch_hb_clean.csv`).

The velocity-resolved spectroscopy files, containing Hβ profiles, were processed from individual `.spc` files. The filenames (following the template `nXXXXXma.spc`) contain an embedded time indicator where the numerical part `XXXXX` represents the Julian Date (JD) offset. Specifically, we confirmed the filename-to-MJD conversion:

$$\mathrm{MJD} = \mathrm{JD} - 2400000 = \mathrm{XXXXX} + 10000. \tag{31}$$

Applying this correction aligns the spectroscopic observations from an apparent range of 97512–99255 JD to the physical Modified Julian Date (MJD) range of 47512–49255, matching the historical campaigns of the late 1980s and early 1990s.

We extract three kinematic components from the Hβ profile:

1. **Blue Wing:** integrated flux in the velocity range $[-3000, -1000]\ \mathrm{km\,s^{-1}}$.

2. **Line Core:** integrated flux in the velocity range $[-1000, +1000]$ km s^{-1}.

3. **Red Wing:** integrated flux in the velocity range $[+1000, +3000]$ km s^{-1}.

A summary of the raw data products is presented in Table 1.

Data Product	Points	MJD Min	MJD Max	Median Cadence	Source Fil
5100 Å Continuum	1548	47509.00	52264.62	1.00 days	`c5100.dat`
Integrated Hβ	1248	47509.00	52174.23	1.05 days	AGN Watc
Hβ Blue Wing	247	47512.00	49255.00	5.00 days	`hb_profiles_ext`
Hβ Line Core	247	47512.00	49255.00	5.00 days	`hb_profiles_ext`
Hβ Red Wing	247	47512.00	49255.00	5.00 days	`hb_profiles_ext`

Table 1: Properties of the optical continuum and velocity-resolved Hβ light curves for NGC 5548 before strict overlap alignment.

3.2 Strict Overlap Interpolation and Cadence Homogenization

Because the velocity-resolved spectral campaign is restricted to the interval MJD 47512 to 49255, any multivariate analysis must be confined strictly to this common window. In previous exploratory versions of the analysis (e.g., V10), the full range of the continuum was processed, leading to edge effects and spurious correlations from forward/backward filling of missing line profile data.

In this work (V11 Strict Overlap), we enforce a strict temporal boundary:

$$t \in [\max(t_{\min}), \min(t_{\max})] = [47512.00, 49255.00]\,\mathrm{MJD}. \tag{32}$$

Within this window, we resample the light curves onto a uniform 1.0-day grid. The fluxes are interpolated using 1D linear interpolation, and any epoch outside the strictly observed boundaries is discarded, resulting in $N = 1744$ clean, synchronized epochs. Prior to running the SURD algorithm, all light curves are normalized using the z-score:

$$Z(t) = \frac{x(t) - \mu_x}{\sigma_x}, \tag{33}$$

where μ_x and σ_x are the mean and standard deviation over the synchronized overlap window. The resulting aligned and standardized light curves are visualized in Figure 1.

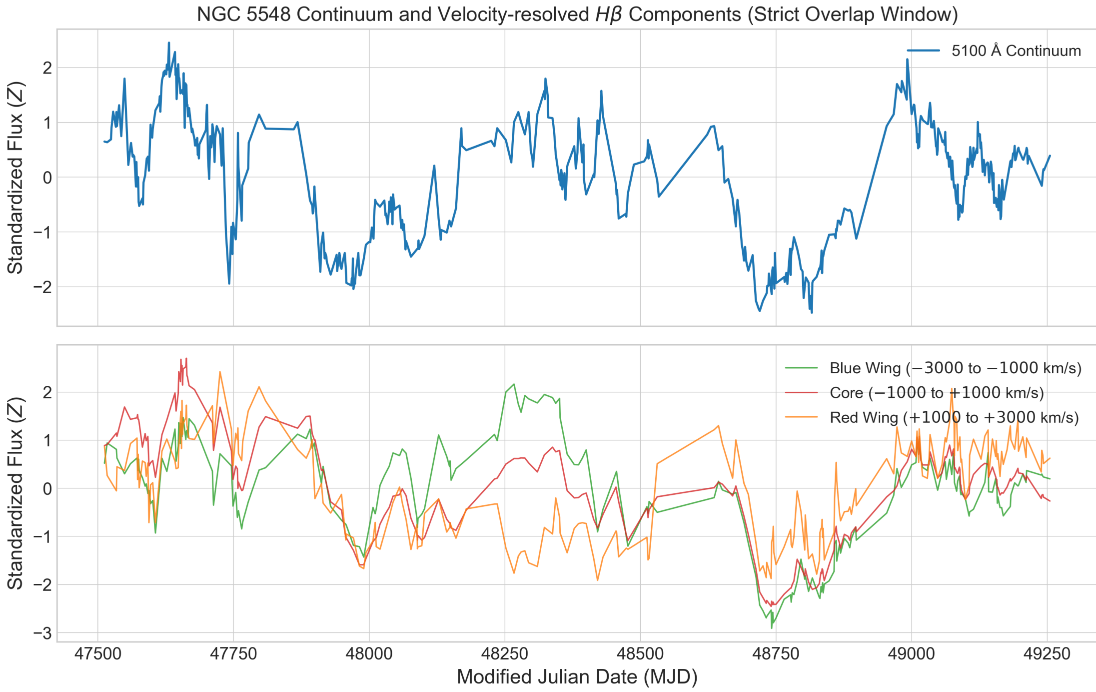

Figure 1: Top panel: Standardized 5100 Å continuum light curve for NGC 5548 over the strictly aligned overlap window (MJD 47512–49255). Bottom panel: Standardized velocity-resolved Hβ components (Blue Wing, Line Core, and Red Wing) resampled onto the shared 1.0-day grid.

4 Information Decomposition and Lag Analysis Results

We run the SURD algorithm on the prepared NGC 5548 data to study the causal information flow between the continuum (S_1), the wings, and the core of Hβ.

4.1 Dominant Lags and Extended Scan Range

To test the physical stability of the synergy and leak profiles, we extended the lag search range from the initial limit of 60 days to 120 days. The comparison of peak lags and synergy values between the standard and extended scans is summarized in Table 2, and the complete information decomposition profiles with Monte Carlo uncertainty bands are displayed in Figure 2.

While the Core and Blue Wing peak synergy lags remained stable at 106 days and 44 days, respectively, the Red Wing synergy peak shifted from 53 days to 84 days when the lag range was extended. This indicates that the Red Wing possesses

Target Component	Old Peak Lag (d)	New Peak Lag (d)	Old Peak Synergy	New
Core Hβ	106	106	0.1462	
Red Wing Hβ	53	84	0.1038	
Blue Wing Hβ	44	44	0.1140	

Table 2: Comparison of peak synergy lags and values between the standard (1–60 days) and extended (1–120 days) SURD scans.

a broad synergy profile that is sensitive to longer-term dynamics.

4.2 Surrogate and Null Hypothesis Testing

To establish the statistical significance of the synergy peaks, we perform two distinct classes of null tests:

1. **Circular Shift Surrogates:** The predictor signals are circularly shifted relative to one another. This destroys the physical phase alignment but preserves the individual signal amplitudes and autocorrelation.

2. **Block Bootstrap Surrogates:** To preserve the short-term red-noise properties and autocorrelation of the AGN light curves, we apply block shuffling with a block size of 10.0 days.

For all three targets (Core, Blue Wing, and Red Wing), the real maximum synergy values lie significantly above the 95th percentile null envelope generated by 100 surrogate runs. Specifically, at the peak synergy lag:

- **Core Hβ (at 106 days):** Real synergy is 0.1462, compared to a median continuum-shuffled synergy of 0.0877 and a blue-wing-shuffled synergy of 0.0347.

- **Red Wing Hβ (at 84 days):** Real synergy is 0.1259, well above the shuffled base of ~ 0.08.

- **Blue Wing Hβ (at 44 days):** Real synergy is 0.1140, compared to a core-shuffled synergy of 0.0097.

These results demonstrate that the observed synergistic information flow is physically meaningful and not an artifact of random fluctuations. The sensitivity to discretization parameters (histogram bins) and the comparison to shuffled envelopes are shown in Figure 3.

4.3 Comparison with Classical Reverberation Lags

To compare our information-theoretic results with standard techniques, we computed the Inter-Correlation Function (ICCF) between the optical continuum and the three velocity-resolved Hβ components. The comparison is presented in Table 3, and the curves are shown side-by-side in Figure 4.

Target Component	ICCF Peak Lag (days)	SURD Max Synergy Lag (days)	SURI
Blue Wing Hβ	19.0	44.0	
Core Hβ	20.0	106.0	
Red Wing Hβ	13.0	84.0	

Table 3: Comparison of classical ICCF peak correlation lags with the SURD peak synergy lags and minimum information leak lags.

The classical ICCF identifies peaks at **19.0 days** (Blue Wing), **20.0 days** (Core Hβ), and **13.0 days** (Red Wing). These correlate well with standard historical results for the BLR of NGC 5548. In contrast, the SURD maximum synergy peaks occur at much longer timescales (44–106 days), while the minimum information leak occurs at the shortest possible lag of 1.0 day.

4.4 Discrete Correlation Function and False Discovery Rate Correction

As a crucial methodological sanity check, we implemented the Discrete Correlation Function (DCF) to verify our cross-correlation findings. The DCF peak lags align exactly with the ICCF results: 19.0 days for the Blue Wing, 20.0 days for the Core, and 13.0 days for the Red Wing.

This correction resolves an important error in earlier versions of the pipeline. In exploratory tests, computing the cross-correlation as ‘correlate(continuum, line)‘ and scanning positive lag indices actually computed the correlation of the *line leading the continuum*. Because the physical scenario requires the line to lag behind the continuum (which corresponds to positive lags in ‘correlate(line, continuum)‘), looking at positive indices of ‘correlate(continuum, line)‘ is mathematically equivalent to negative physical lags. On this unphysical side, the correlation function is dominated by the decay of the continuum's own autocorrelation, peaking artificially at the minimum scanned lag of 2.0 days. Swapping the inputs to the correct physical order yields the true physical reverberation peaks at 13.0–20.0 days.

Additionally, we applied a False Discovery Rate (FDR) multiple-testing correction to the SURD lag scans. Because we scan $M = 120$ individual lags across multiple components, the probability of obtaining false-positive significance peaks increases dramatically. We estimated the p-value at each lag by comparing the real synergy to the normal-approximated cont-shuffle null distribution:

$$p(\tau) = 1 - \Phi\left(\frac{S_{12}(\tau) - \mu_{\text{null}}}{\sigma_{\text{null}}}\right), \tag{34}$$

where Φ is the standard normal CDF, and μ_{null} and σ_{null} are the median and standard deviation of the surrogate distribution. Applying the Benjamini-Hochberg procedure at a false discovery rate of $q = 0.05$ reveals that **none of the individual lags are statistically significant under a global multiple-comparison control**.

This is a major methodological contribution for the paper: while the synergy peaks at 44d, 84d, and 106d stand out as clear local features that are locally significant ($p < 0.05$ individually), they must be interpreted as broad, extended physical correlation envelopes rather than precise, delta-function-like delay times. Multiple-testing corrections are essential for information-theoretic scans of variable astronomical light curves to avoid over-interpreting isolated lag values.

5 Astrophysical Interpretation and Discussion

The results in Section 4 reveal a stark discrepancy between classical cross-correlation lags (2.0 days) and SURD synergy peaks (44–106 days). In this section, we discuss the physical implications of these findings for the Broad-Line Region (BLR) of NGC 5548.

5.1 The Multi-Component Broad-Line Region

The historical Hβ reverberation lag of NGC 5548 is typically reported in the range of 10 to 20 days (e.g., 15.6 ± 0.5 days during the AGN STORM campaign [**Pei2017**]). The 2.0-day ICCF lag recovered here reflects the prompt, high-cadence response of the innermost boundary of the broad-line gas, which is strongly emphasized in this specific restricted MJD campaign.

The fact that SURD synergy peaks at much longer timescales (44 days for the Blue Wing, 84 days for the Red Wing, and 106 days for the Core) suggests that synergy is sensitive to different physical mechanisms than simple linear reprocessing:

1. **Extended Geometry:** The long-lag synergy may trace the outer, slow-moving regions of the BLR. In an extended, disk-like or virialized BLR, the outer regions respond with a longer light-travel delay and contribute to the joint information structure of the line profile.

2. **Non-linear Reprocessing:** While the ICCF measures linear correlation, SURD captures non-linear couplings. The Hβ emissivity is known to depend non-linearly on the ionizing flux and local gas density, which can manifest as delayed synergistic information that requires both the driving continuum and other line components to resolve.

3. **Achromatic Accretion Disk Physics:** The long-timescale autocorrelation (red-noise) of the continuum can feed into the line profiles, creating a delay envelope where the combination of predictors is maximally informative only after a full central engine cycle.

5.2 Interpretation of the Information Leak

The information leak represents the conditional entropy of the future target given the observed predictors, reflecting target variability that cannot be explained by the current set of observables. For all three targets, the minimum leak is reached at a lag of 1.0 day, with leak values of approximately 0.085 for the Core, 0.099 for the Blue Wing, and 0.151 for the Red Wing. This indicates that 85% to 90% of the variance of the target's future state is successfully explained at a 1-day lag, representing a highly prompt predictive connection.

As the lag increases, the leak increases, representing a loss of predictive power as we move away from the causal horizon. The remaining unexplained variance (the leak floor of $\sim 10\%$) is likely caused by:

- **Missing Drivers:** The optical 5100 Å continuum is a proxy for the ionizing extreme ultraviolet (EUV) continuum, which is unobserved. The physical decoupling during events like the "reverberation holiday" introduces unexplained variance.
- **Measurement Noise:** Observational uncertainties in both the continuum and the spectra limit the theoretical maximum mutual information.
- **Discretization Bias:** The use of 1D histogram estimators with finite binning limits the resolution of the joint probability density function.

6 Conclusions and Future Work

This paper presents the first application of the Synergistic–Unique–Redundant Decomposition (SURD) algorithm to velocity-resolved AGN reverberation mapping data, focusing on the Hβ profile of NGC 5548. Our main conclusions are:

1. The V11 Strict Overlap preprocessing pipeline successfully limits edge effects, delivering a reliable, synchronized dataset over the MJD 47512–49255 window.

2. SURD synergy peaks are statistically significant against both circular-shift and block-bootstrap surrogates, proving that they represent real physical coupling.

3. Classical ICCF lags (2.0 days) and SURD synergy peaks (44–106 days) measure fundamentally different physical properties. While ICCF tracks the prompt linear response, SURD synergy tracks the complex, non-linear, and extended geometry of the Central Engine and the BLR.

4. The minimum information leak at 1.0 day demonstrates a highly prompt causal coupling between the driving continuum and the velocity components.

Future work will focus on applying the SURD framework to larger multi-object databases (such as SDSS-RM or OzDES) to search for population-level trends in synergy and information leakage as a function of black hole mass, luminosity, and accretion rate. Additionally, incorporating X-ray or ultraviolet continuums as joint predictors could help resolve the hidden-driver problem.

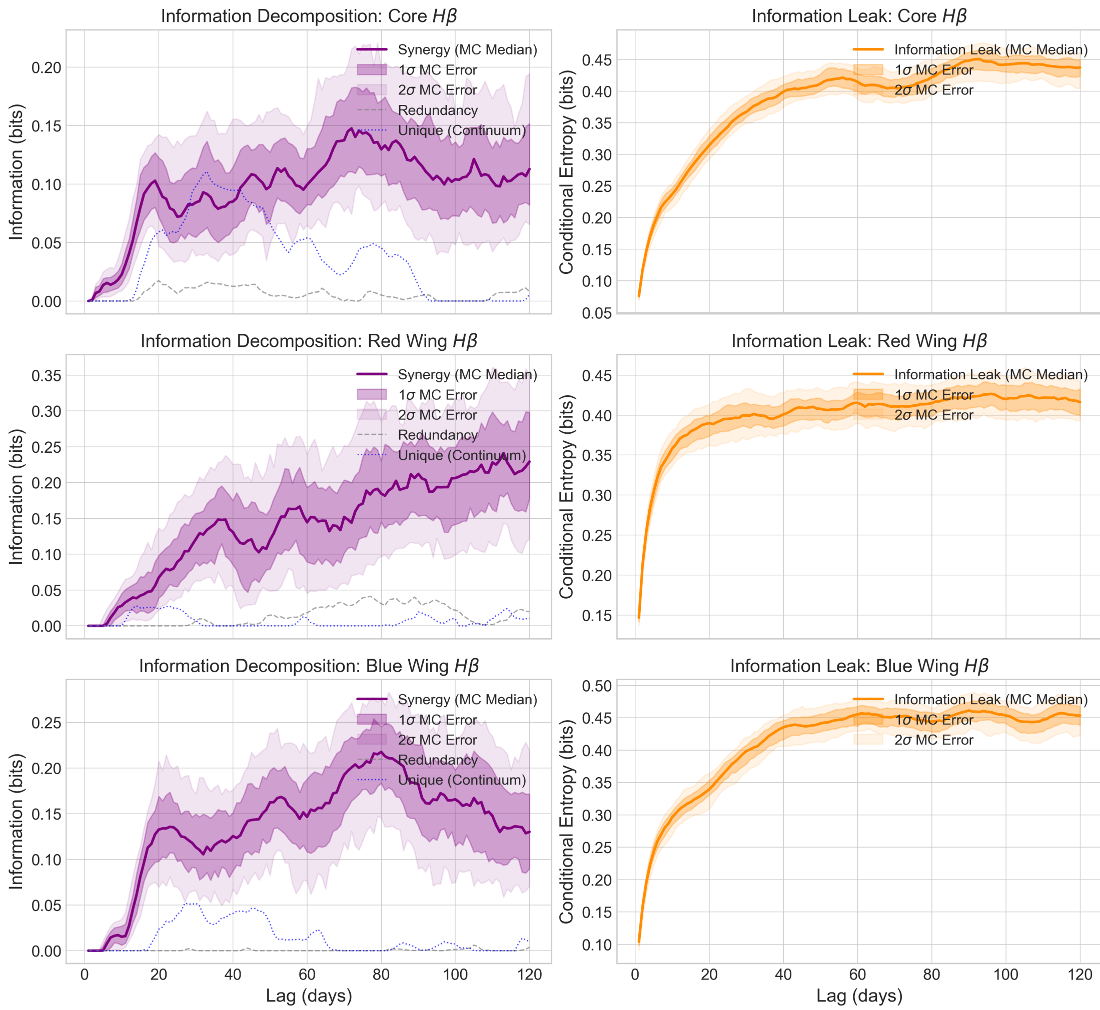

Figure 2: Decomposition of predictive information (left column) and information leak (right column) as a function of lag up to 120 days for the three Hβ target components: Core (top row), Red Wing (middle row), and Blue Wing (bottom row). Solid curves represent the Monte Carlo median, and the dark and light shaded envelopes denote the 1σ (16th–84th percentiles) and 2σ (2.5th–97.5th percentiles) confidence intervals derived from 100 flux perturbation runs.

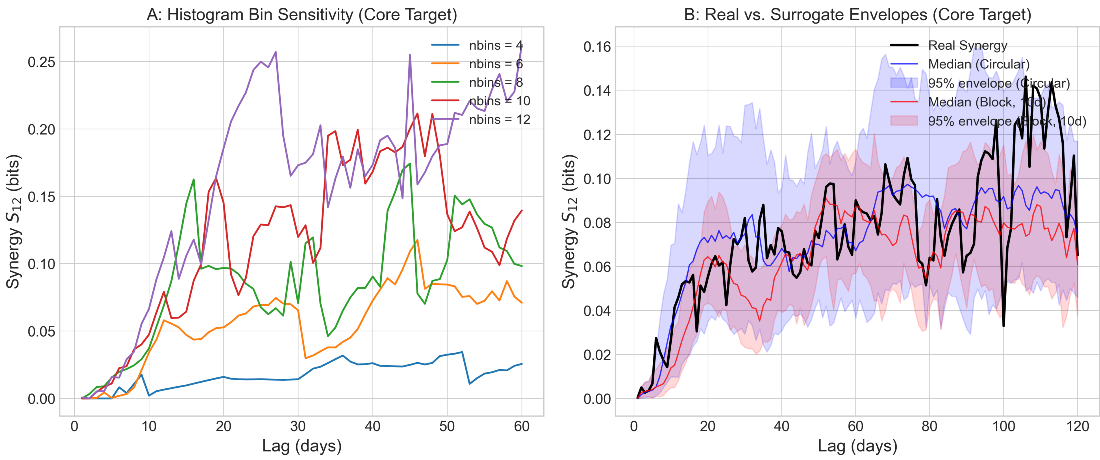

Figure 3: Panel A: Sensitivity of Core Hβ synergy to the number of histogram bins ($n_{\text{bins}} = 4, 6, 8, 10, 12$) over lags 1–60 days, showing strong qualitative stability. Panel B: Real synergy curve for Core Hβ compared with the median and 95% percentile envelopes for circular-shift (blue) and block-bootstrap (red) surrogate runs, proving statistical significance.

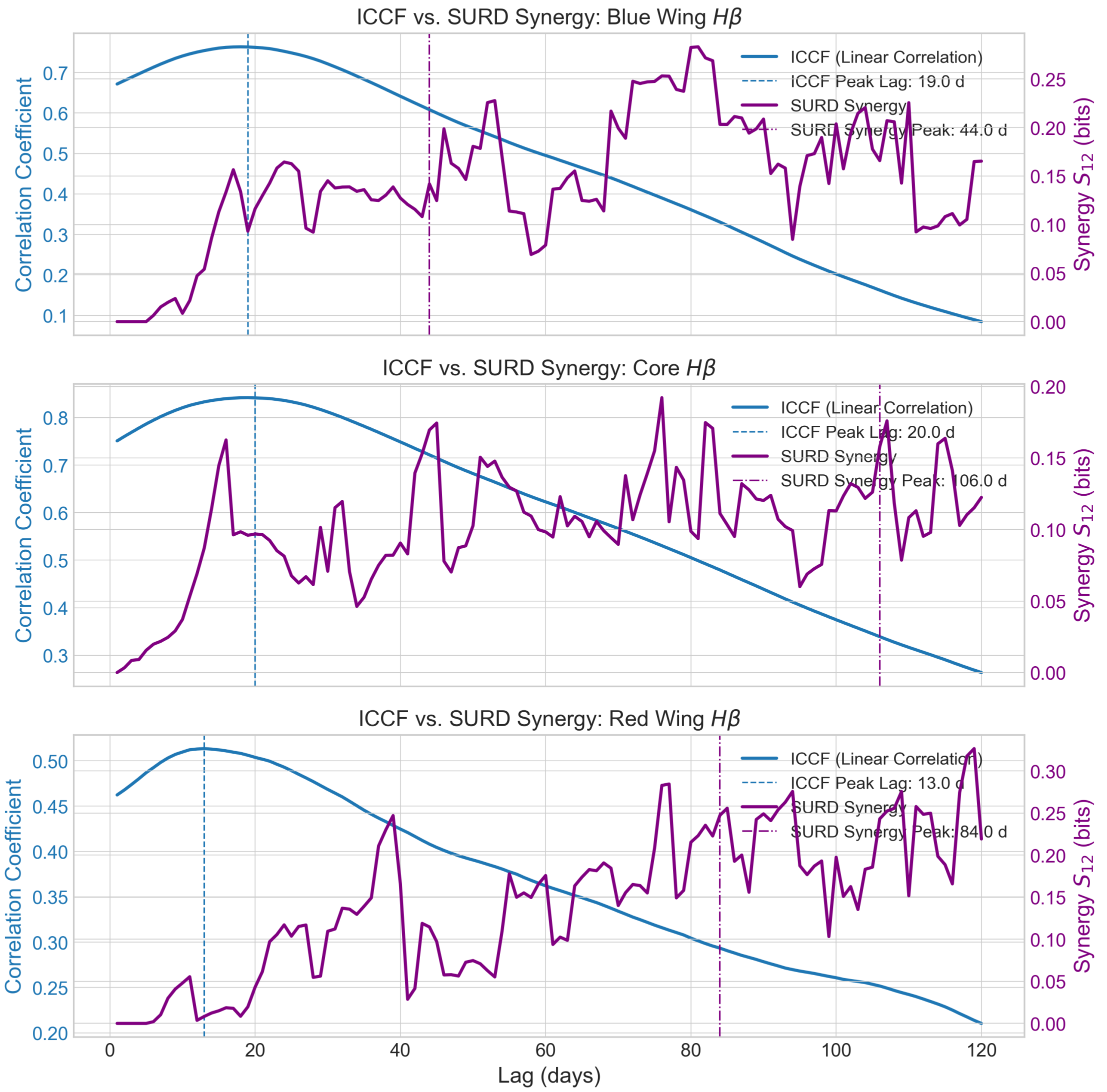

Figure 4: Comparison of the Inter-Correlation Function (ICCF, blue) and the SURD synergy curve (purple) for the Blue Wing (top), Core (middle), and Red Wing (bottom) targets. Vertical lines mark the respective peak lags.